Digital Signal Processing 2025 project

Sub-band Audio Coding Using a Two-Band FIR Filter Bank

Author: Vaibhav Khute

Abstract

In this project, a simple sub-band audio coding system is designed and analyzed using digital signal processing techniques. The main objective is to reduce the bitrate of a speech signal while preserving acceptable audio quality. The system uses a two-band FIR filter bank to split the signal into low-frequency and high-frequency components. Each band is then downsampled and quantized with different bit depths before reconstruction. The performance of the system is evaluated using time-domain and frequency-domain analysis, spectrograms, and objective metrics such as Mean Squared Error (MSE), Signal-to-Noise Ratio (SNR), and compression ratio.

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1. Introduction and Motivation

Digital audio signals require a large amount of data for storage and transmission. For example, an uncompressed mono speech signal sampled at 8 kHz with 16-bit resolution requires a bitrate of 128 kbps. In many practical applications, such as communication systems, it is desirable to reduce this bitrate while maintaining speech intelligibility.

Sub-band audio coding is a classical DSP approach that takes advantage of the frequency characteristics of audio signals. Speech signals typically contain most of their energy in the low-frequency range, while high-frequency components contribute less to perceived quality. By splitting the signal into frequency bands and processing each band separately, it is possible to achieve efficient compression.

The motivation of this project is to gain practical experience with sub-band signal processing and to study the trade-off between compression efficiency and audio quality using a simple and understandable system.

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2. Theoretical Background

2.1 Sampling and Quantization

Sampling converts a continuous-time signal into a discrete-time signal by taking samples at a fixed sampling frequency. According to the Nyquist sampling theorem, the sampling frequency must be at least twice the maximum signal frequency to avoid aliasing.

Quantization maps continuous-valued samples to a finite number of discrete amplitude levels. Reducing the number of quantization bits lowers the bitrate but introduces quantization noise, which affects signal quality.

2.2 Sub-band Signal Processing

Sub-band processing divides a signal into multiple frequency bands using digital filters. Each band can then be processed independently. In this project, a two-band system is used:

- A low-pass filter extracts low-frequency components
- A high-pass filter extracts high-frequency components

This separation allows different sampling rates and quantization resolutions to be applied to different frequency regions.

2.3 FIR Filter Design

Finite Impulse Response (FIR) filters are used due to their stability and linear phase properties. The filters are designed using the window method with a Hamming window. The cutoff frequency is chosen as one quarter of the sampling frequency, which allows safe downsampling by a factor of two without aliasing.

2.4 Downsampling and Upsampling

Downsampling reduces the sampling rate by keeping every second sample, which reduces the data rate. Before downsampling, filtering is required to remove frequency components that could cause aliasing. Upsampling restores the original sampling rate by inserting zeros between samples and applying synthesis filters.

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3. System Description and Methodology

The implemented sub-band audio coding system consists of the following steps:

1. Input speech signal
2. Analysis filter bank (low-pass and high-pass FIR filters)
3. Downsampling of each sub-band by a factor of two
4. Uniform quantization with different bit depths for each band
5. Upsampling of each sub-band
6. Synthesis filter bank and signal reconstruction

The low-frequency band is allocated more quantization bits than the high-frequency band, since it carries more perceptually important information.

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4. Experimental Setup

4.1 Input Signal

The input signal used in this project is a mono speech recording with a sampling frequency of 8 kHz and a duration of approximately one minute. The signal is normalized prior to processing to ensure numerical stability.

4.2 Frequency Analysis of the Original Signal

Before applying sub-band processing, the signal is analyzed in the frequency domain using the Fast Fourier Transform (FFT) and spectrogram.

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Sampling frequency: 8000
Signal length (samples): 466063
Duration (seconds): 58.257875
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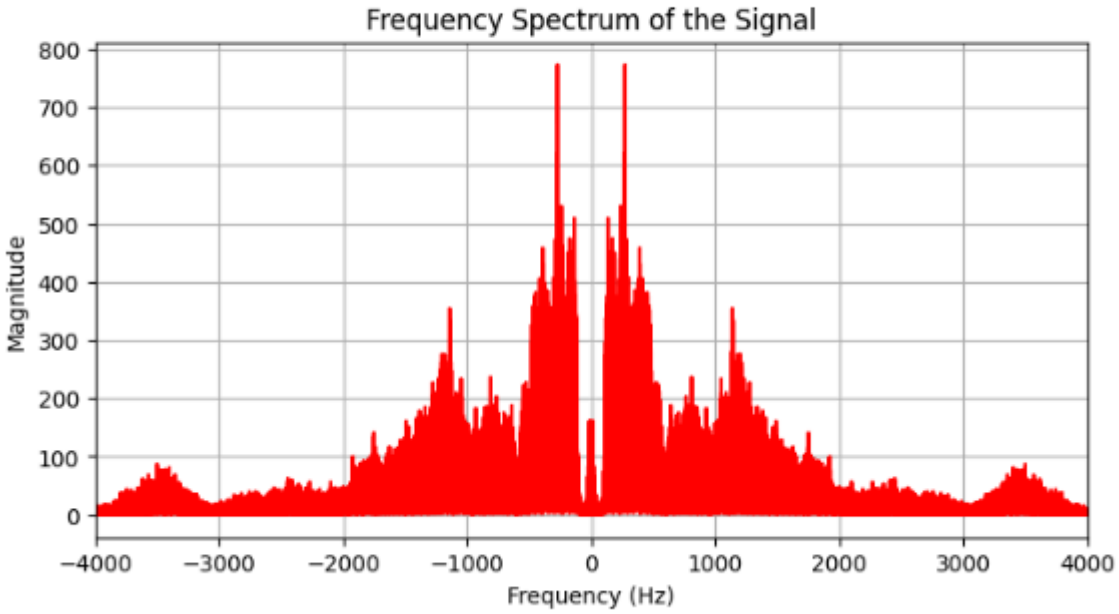

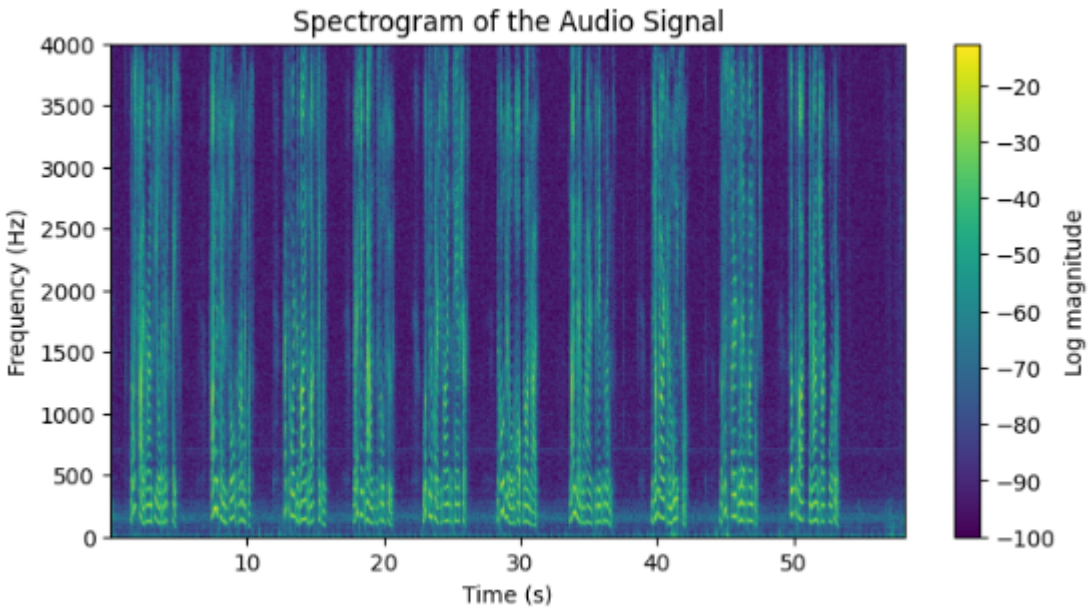

These plots show that most of the signal energy is concentrated at low frequencies, which motivates the use of sub-band coding.

4.3 Filter Design

Low-pass and high-pass FIR filters are designed using the window method. The cutoff frequency is set to fs/4 to allow downsampling by a factor of two.

[Insert Figure 4: Frequency response of the low-pass and high-pass FIR filters]

4.4 Sub-band Processing and Quantization

Each filtered sub-band is downsampled by a factor of two. Uniform quantization is applied with different bit depths:

- Low-frequency band: higher number of bits
- High-frequency band: lower number of bits

Different bit allocations are tested to study their effect on audio quality and compression.

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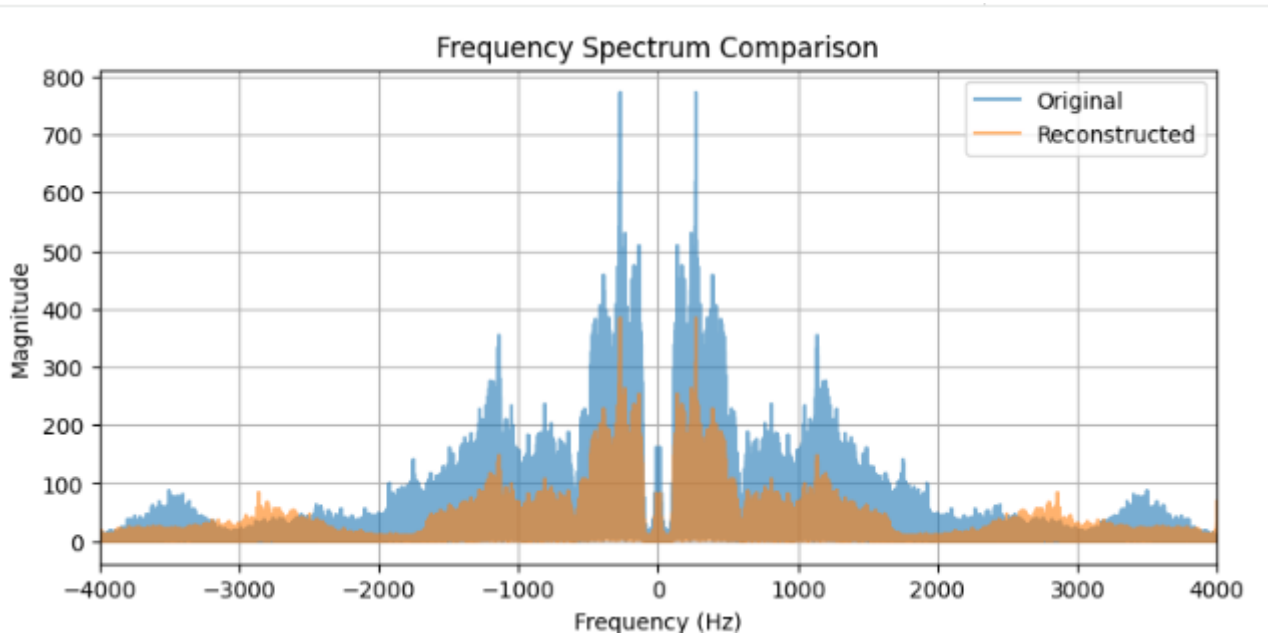
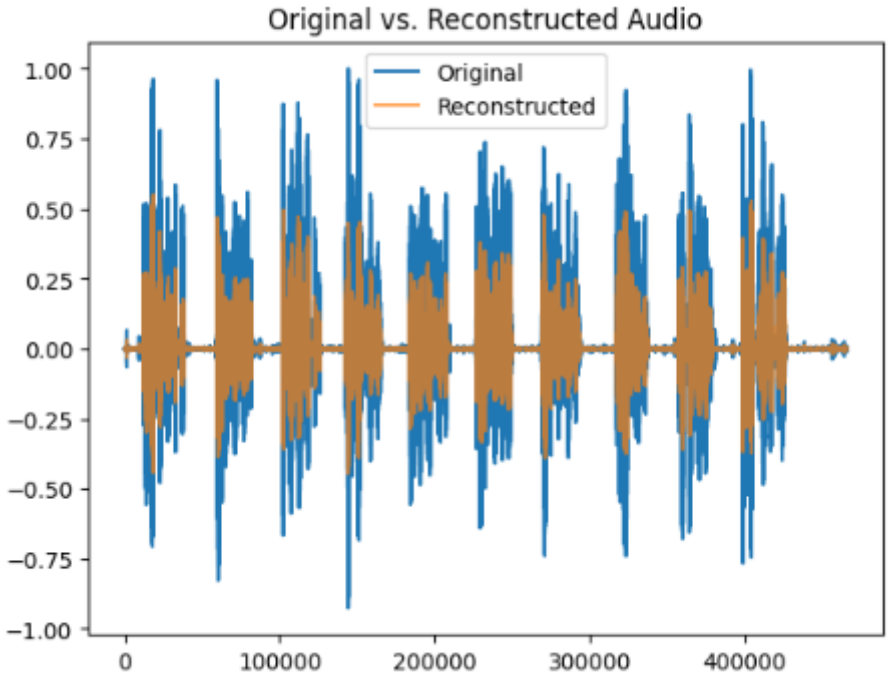
5. Reconstruction and Evaluation

After quantization, each sub-band is upsampled and passed through synthesis filters. The reconstructed signal is obtained by summing the reconstructed sub-bands.

5.1 Time-domain Comparison

The reconstructed signal is compared with the original signal in the time domain.





The reconstructed signal closely follows the original waveform, with small differences caused by quantization and filtering.

5.2 Frequency-domain Comparison

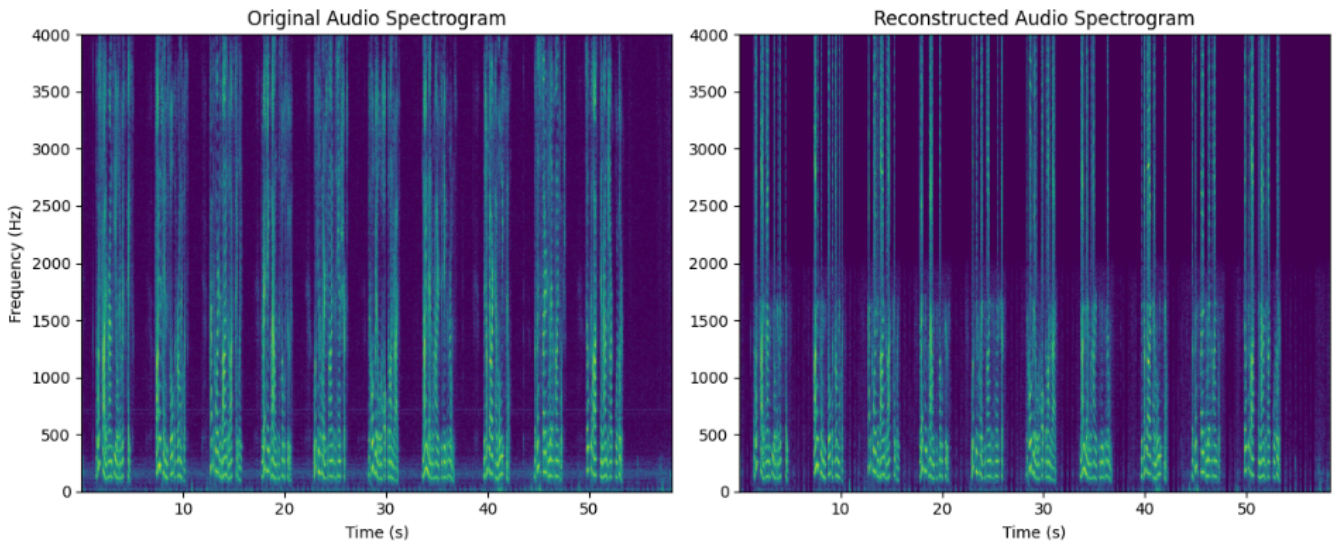
The frequency spectra of the original and reconstructed signals are compared.

[Insert Figure 7: Frequency spectrum comparison of original and reconstructed signals]

The reconstructed signal shows reduced high-frequency energy and a slightly increased noise floor, which are expected effects of compression.

5.3 Spectrogram Comparison

Spectrograms are used to compare the time–frequency content of the original and reconstructed signals.



The reconstructed spectrogram shows smoother structures and reduced high-frequency detail compared to the original signal.

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6. Objective Performance Metrics

The system performance is evaluated using the following metrics:

- Mean Squared Error (MSE)
- Signal-to-Noise Ratio (SNR)
- Bitrate and compression ratio

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Mean Squared Error (MSE): 0.010104
Signal-to-Noise Ratio (SNR): -0.62 dB
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```
Original Bit Rate: 128.0 kbps
Compressed Bit Rate: 48.0 kbps
Compression Ratio: 2.67
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The original signal has a bitrate of 128 kbps. After sub-band coding, a compressed bitrate of approximately 48 kbps is achieved, resulting in a compression ratio of about 2.6.

It is observed that reducing the number of bits in the low-frequency band significantly degrades SNR, while changes in the high-frequency band have a smaller effect.

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7. Discussion

The experimental results demonstrate the effectiveness of sub-band audio coding for speech signals. The system achieves significant bitrate reduction while maintaining intelligible audio quality. The results confirm that low-frequency components are more important for perceived speech quality and should be allocated more bits.

The trade-off between compression and quality is clearly visible in both objective metrics and visual analysis. More aggressive compression leads to increased distortion, while higher bit depth improves quality at the cost of reduced compression.

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8. Conclusion

In this project, a two-band sub-band audio coding system using FIR filters was successfully implemented and evaluated. The system combines filtering, downsampling, and quantization to reduce audio bitrate while preserving speech intelligibility.

The project provided practical insight into key DSP concepts such as digital filtering, sub-band processing, and quantization. Future work could extend the system to more sub-bands or incorporate perceptual models for improved compression performance.